SCOPE-Fin: Budget-Aware Orchestration of Retrieval, Quantitative Reasoning, and Risk Triage for Financial Research

A Selective-Routing Framework and Evaluation Protocol

Xiang Zhang
College of Computer Science, Sichuan University
2024324022023@stu.scu.edu.cn
October 2026

Abstract

Financial research assistants face heterogeneous requests: some need a single fact lookup, others require multi-document synthesis, numerical calculation, or temporal transaction analysis. Applying the most expensive model to every request wastes computation, while routing every request through a lightweight model risks inaccurate or unsupported answers. This paper proposes SCOPE-Fin (Selective Cost-aware Orchestration for Processing Evidence in Finance), a framework that selects specialist reasoning routes according to task type, expected information gain, evidence quality, and a per-request budget. Its route library combines direct retrieval, multi-hop financial RAG, symbolic numerical execution, and temporal risk scoring; a quality-cost gate either accepts a supported draft, invokes another route, or escalates the case for analyst review. The design synthesizes ideas from three supplied papers on blockchain anomaly detection, financial document QA, and modular mathematical reasoning, but shifts the research question from evidence integration to selective allocation of computation and human attention. We formalize the routing objective, define a benchmark-based evaluation protocol, and state testable hypotheses concerning answer quality, cost, and review burden.

Keywords: financial research; adaptive routing; computational budget; retrieval-augmented generation; symbolic reasoning; human-AI collaboration; selective prediction

1 Introduction

Financial research systems are asked to perform tasks with very different computational demands. Looking up a dated disclosure may require only targeted retrieval, while reconciling values across a report may need multiple retrieval hops and executable arithmetic. A transaction-risk question can instead require temporal modeling over a sequence of events. Treating all of these requests as one generic text-generation problem can spend computation on easy cases and under-serve difficult ones.

Retrieval-augmented generation (RAG) provides a way to condition model output on external documents [5]. Financial QA benchmarks such as FinQA and TAT-QA illustrate that answering questions over reports can involve both numerical operations and mixed text-table evidence [6, 7]. Meanwhile, blockchain transaction analysis involves different inputs and objectives: methods such as the Elliptic graph-learning work and the uploaded GAM-CoT paper focus on identifying suspicious temporal or relational patterns [4, 1]. A single pipeline is unlikely to be the most economical route for every task.

This paper proposes SCOPE-Fin, a selective orchestration framework for a financial research assistant. It uses a task profiler to estimate the request type and difficulty, a policy to choose a subset of specialist routes, and a quality-cost gate to decide whether to stop, spend more computation, or request human review.

2 Background and Motivation

2.1 Specialist components in financial AI: The GAM-CoT paper describes feature- and time-aware attention for transaction anomaly detection. Its target is a structured event sequence, not a general document question [1]. The FinLLaMA-RAG preprint instead describes dense retrieval, iterative multi-hop reasoning, and context aggregation for document-level financial QA [2]. MetaMath-LLaMA proposes a modular scheduler and a symbolic-neural computation path for interpretable multi-step mathematics [3].

2.2 Selective prediction and human review: Selective prediction studies systems that may withhold an answer when confidence is insufficient, trading answer coverage against risk [11]. Human-AI interaction guidance emphasizes keeping people informed about system limitations and providing a way to correct or override automated behavior [10]. SCOPE-Fin applies these ideas operationally: an answer is not accepted solely because a language model is confident; route evidence, calculation checks, and task-specific validation contribute to the stopping decision.

2.3 Source-reported ablation evidence: The supplied FinLLaMA-RAG preprint reports a full-model ablation table with four metrics. Its table lists nDCG@10 of 0.62, BLEU of 30.5, ROUGE-L of 35.2, and F1 of 0.75 for the full system; its retrieval-only and generation-only rows report lower scores on each listed metric.

3 Problem Definition

Let a request be q, available evidence be x, and remaining computation budget be B. SCOPE-Fin has a finite route set R = {r0, r1, r2, r3}: direct evidence lookup, multi-hop document QA, symbolic quantitative reasoning, and temporal transaction-risk analysis.

A policy pi maps the request and current state to the next action: invoke a route, return a draft, or escalate to an analyst. We define the objective as: pi* = arg max_pi E_{q,x}[U(y, q, x) - lambda_c C(pi) - lambda_h H(pi)] subject to C(pi) <= B, Runsupported(pi) <= delta, where U is task utility, C is computational cost, H is human-review effort, and Runsupported is the rate of unsupported claims among accepted answers.

4 SCOPE-Fin Architecture

4.1 Task profiler and route library: The task profiler extracts a compact feature vector psi(q, x): requested output type, presence of numeric operators, number of apparent source entities, temporal span, retrieval ambiguity, and impact category. It assigns a provisional route distribution, not a final answer.

Route Library includes: (1) Direct lookup: one fact, known source, low ambiguity, low cost. (2) Multi-hop document QA: cross-section or cross-document synthesis, medium to high cost. (3) Symbolic quantitative reasoning: arithmetic, comparison, unit conversion, or constraint check, low to medium cost. (4) Temporal risk analysis: ordered transaction or event sequence with risk labels, medium to high cost. (5) Analyst escalation: conflicting sources, insufficient support, or high-impact uncertainty.

4.2 Value-of-computation routing: For candidate route r at state zt, let Delta U(r | q, zt) estimate expected improvement in task utility, Delta C(r) estimate incremental computational cost, and Delta H(r) estimate the effect on review effort. The score is V(r | q, zt) = Delta U - lambda_c Delta C - lambda_h Delta H.

5 Evaluation Protocol and Hypotheses

The evaluation separates route quality from route-selection quality. Elliptic is used for temporal transaction-risk tasks, while FinQA and TAT-QA are used for financial numerical and text-table QA [4, 6, 7]. A new mixed-request test set is needed to evaluate routing decisions across task types.

Hypothesis H1: At a pre-specified non-inferiority margin, selective routing uses less computation than the always-on policy while preserving task quality.

Hypothesis H2: Routing numerical questions to symbolic execution reduces arithmetic error versus direct generation.

Hypothesis H3: Escalation improves selective accuracy at matched answer coverage on mixed-request tasks.

6 Discussion, Limitations and Conclusion

SCOPE-Fin treats computation as an allocatable resource rather than an unlimited background service. The approach is most useful where requests vary substantially in difficulty and where a human analyst can review the uncertain tail.

Limitations: Route-value estimates can be miscalibrated, and a policy may systematically under-allocate effort to unfamiliar or minority cases. Benchmarks do not capture every reporting convention, jurisdiction, or analyst workflow.

Conclusion: This paper proposed SCOPE-Fin, a budget-aware routing framework for financial research tasks. Selective orchestration should be evaluated on the quality-cost frontier and on human review burden, rather than presumed effective because specialist components perform well in isolation.

References

[1] X. Huang, C. Zhao, X. Li, C. Feng, and W. Zhang, "GAM-CoT Transformer: Hierarchical Attention Networks for Anomaly Detection in Blockchain Transactions," Journal of Emerging Applied Artificial Intelligence, 2025.

[2] X. Huang, Z. Lin, F. Sun, W. Zhang, K. Tong, and Y. Liu, "Enhancing Document-Level Question Answering via Multi-Hop Retrieval-Augmented Generation with LLaMA 3," TechRxiv preprint, 2025.

[3] X. Huang, Z. Wang, X. Liu, Y. Tian, and Q. Leng, "Towards Interpretable and Consistent Multi-Step Mathematical Reasoning in Large Language Models," manuscript provided by the authors, 2025.

[4] M. Weber, G. Domeniconi, J. Chen, et al., "Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics," arXiv:1908.02591, 2019.

[5] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," NeurIPS, 2020.

[6] Z. Chen et al., "FinQA: A Dataset of Numerical Reasoning over Financial Data," EMNLP, pp. 3697–3711, 2021.

[7] F. Zhu et al., "TAT-QA: A Question Answering Benchmark on a Hybrid of Tabular and Textual Content in Finance," ACL-IJCNLP, pp. 3277–3287, 2021.

[8] J. Wei et al., "Chain-of-Thought Prompting Elicits Reasoning in Large Language Models," NeurIPS, 2022.

[9] S. Yao et al., "ReAct: Synergizing Reasoning and Acting in Language Models," ICLR, 2023.

[10] S. Amershi et al., "Guidelines for Human-AI Interaction," CHI, pp. 1–13, 2019.

[11] Y. Geifman and R. El-Yaniv, "Selective Classification for Deep Neural Networks," NeurIPS, 2017.